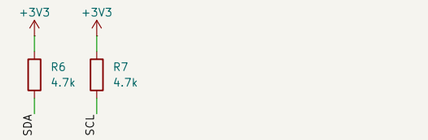
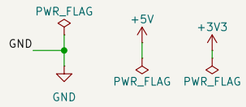
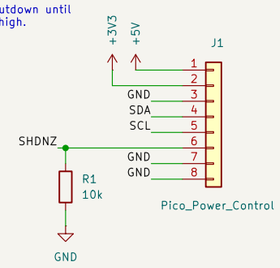
TLV320ADC6140 breakout for Stage 2 of TEST_PLAN_MCU_CAPTURE.md. It connects to a Pico 2 by jumper wires.

POWER AND CONTROL

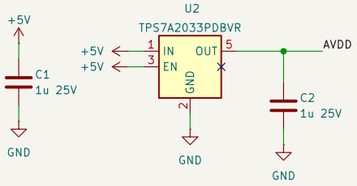


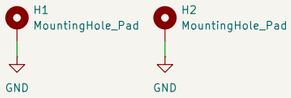
R1 holds the ADC in shutdown until the Pico drives SHDNZ high.

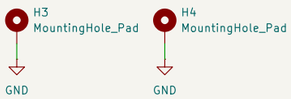




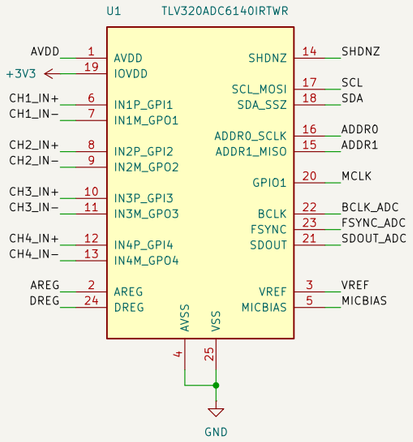
AVDD REGULATOR



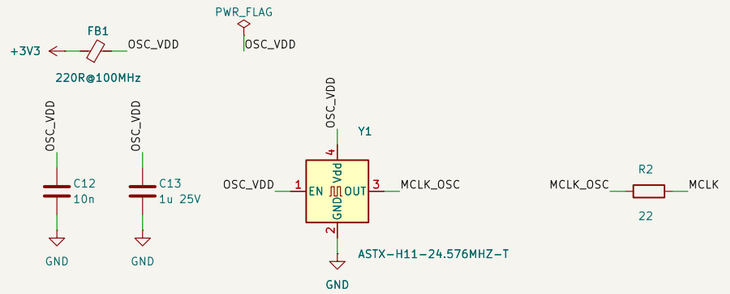




ADC

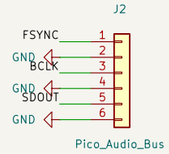


AUDIO REFERENCE 24.576 MHz



Standby pin 1 is tied to VDD, so the TCXO always runs.
MCLK enters the ADC on GPIO1.

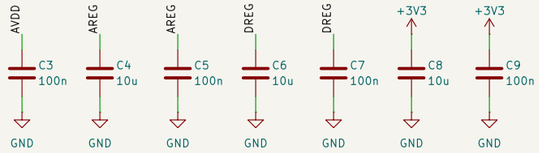
AUDIO BUS (ADC is bus master)

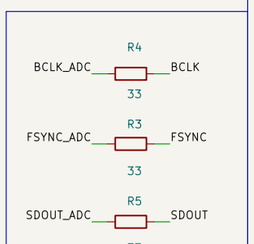


TEST POINTS



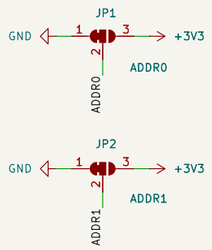
ADC DECOUPLING. Place each 100 nF cap at its pin.





R3–R5 sit next to U1.
Swap for 0 ohm if not needed.

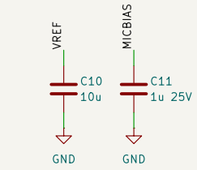
I2C ADDRESS



Default: both bridged to GND, address 0x4C.
Cut 1–2 and bridge 2–3 to change it.

MIC GROUND ISOLATION

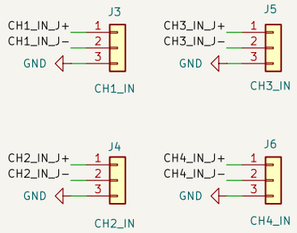
C10 & C11:
Route ground terminal straight to AVSS, no via.



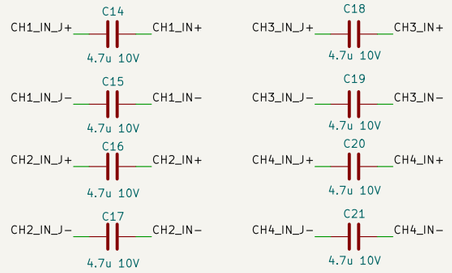
C11:
There are no onboard mics, so MICBIAS is not used.
Populate C11 for decoupling in case of activation in firmware.

ANALOG INPUTS

Header pins:
1 = CHx_IN+ (tip)
2 = CHx_IN− (ring)
3 = GND (sleeve)



ANALOG INPUT COUPLING CAPS



C14–C21 are 4.7 uF X7R, as on Ensor.
Set the ADC inputs to 10 kohm for a 3.4 Hz corner.
With 4.7 uF, firmware must lengthen the input quick-charge time.

Stage 2 test board. Parts follow docs/bom.md, ADC breakout section.

Portable Float Recorder

Sheet: /
File: adc-breakout.kicad_sch

Title: TLV320ADC6140 ADC Breakout

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1